

NICOLÁS ANDRÉS ARAYA CARO

AI Engineer & Computer Vision Specialist

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scholar.google.com/citations?hl=es&user=2H_rhNUAAAAJ

PROFESSIONAL PROFILE

AI Engineer specialized in Artificial Intelligence, Computer Vision, and Machine Learning with doctoral training in Computer Science. Proven experience in applied research (FONDECYT projects), Deep Learning model development, image processing, and multimodal data analysis. Passionate about creating intelligent AI-based solutions for real-world problems.

WORK EXPERIENCE

AI Research Assistant – FONDECYT Project <i>Universidad Diego Portales</i>	Nov 2024 - Dec 2024
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Technical assistant on FONDECYT initiation project N° 11220293 (PHASE 2023)

- Collaboration on research project focused on automating cellular simulators using artificial intelligence
- Development of computational tools for modeling and simulation of complex biological behaviors
- Participation in experimental analysis and validation of computational models
- Technical support in applied research and collaboration with international academic teams

Technologies: Python, PyTorch, Machine Learning, Computational Modelling, Scientific Computing

Part-Time Instructor <i>Universidad Diego Portales</i>	Mar 2023 - Jul 2024
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- Instructor for Probability & Statistics in engineering and computer science programs
- Development of course content in probability, statistics, and quantitative analysis
- Design of evaluations and academic materials focused on problem-solving

Technologies: Teaching, Pedagogy, Quantitative Analysis

AI Research Assistant – FONDECYT Project <i>Universidad Diego Portales</i>	Jun 2023 - Jan 2024
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- Development and evaluation of Machine Learning and Deep Learning models for multimodal classification of biomedical data
- Implementation of data processing, imputation, and analysis pipelines
- Comparative evaluation of RVFL/Deep RVFL algorithms for predictive tasks
- Participation in scientific publications on multimodal AI and predictive analysis

Technologies: Python, Scikit-learn, Deep Learning, Data Processing, Multimodal AI

Computer Vision Engineer <i>Gala Robotics</i>	Jul 2022 - Jan 2023
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- Development of Computer Vision solutions for automated detection of damage on industrial surfaces
- Implementation and evaluation of image processing and visual analysis models

- Collaboration in applied research and experimental results validation

Technologies: Python, OpenCV, Machine Learning, Computer Vision

EDUCATION

Ph.D. in Computer Science (candidate) Pontificia Universidad Católica de Chile	2023
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Research focus: Computer Science

M.Sc. in Engineering Sciences Universidad Diego Portales	2021 - 2022
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Engineering and Technology

B.S. in Computer Engineering & Telecommunications Universidad Diego Portales	2016 - 2022
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Graduated

TECHNICAL SKILLS

Languages & Frameworks: Python (advanced), SQL (advanced), Git (intermediate)

AI & Machine Learning: Deep Learning (advanced), PyTorch (advanced), Scikit-learn (advanced), TensorFlow, Multimodal AI

Computer Vision: OpenCV (advanced), Image Processing, Visual Analysis

Data Science: Pandas (advanced), NumPy (advanced), Data Analysis, Statistical Modelling

Cloud & DevOps: Google Cloud Platform (GCP) (intermediate), Docker (basic), Linux (basic)

Specialization Areas: Applied Artificial Intelligence, Computer Vision, Machine Learning, Computational Modelling, Scientific Computing

LANGUAGES

English: Intermediate

Spanish: Native